

Power BI (Visualizations)

1. Stacked bar chart / Stacked column chart

A **stacked bar chart** visually represents how different components (segments) contribute to a total (each bar).

Properties:

- **Part-to-Whole:** Shows the composition of a total.
- **Two Categorical, One Numeric:** Requires categories for bars and segments, and a numeric value for size.
- **Color-Coded Segments:** Distinguishes components within each bar.
- **Common Baseline:** All bars start from zero for accurate total comparison.
- **Orientation:** Can be vertical (stacked columns) or horizontal (stacked bars).

Uses:

- **Composition Breakdown:** Illustrate how sub-categories make up a total (e.g., sales by product line).
- **Comparative Contribution:** Compare how component proportions vary across different main categories.
- **Trend Tracking (Caution):** Observe changes in total and major component contributions over time.

2. Clustered bar chart / Clustered column chart

A **clustered bar chart** groups multiple bars side-by-side for each main category, allowing direct comparison of individual sub-categories.

Properties:

- **Direct Comparison:** Ideal for comparing specific values among different sub-categories within a group.
- **Two Categorical, One Numeric:** Needs categories for groups, categories for individual bars, and a numeric value.

- **Color-Coded Series:** Distinguishes sub-categories consistently across groups.
- **Common Baseline:** All bars share a baseline for easy value comparison.
- **No Part-to-Whole:** Does not show a sum or composition for the whole group.

Uses:

- **Performance Comparison:** Compare metrics of different items across various groups (e.g., product sales by region).
- **Multiple Series Display:** Show distinct values for several related items per category.

3. 100% Stacked bar chart / 100% Stacked column chart

It's a bar chart where each bar represents a total (often 100%), and is segmented into sub-sections, each representing a percentage contribution of a different category to that total.

Properties:

- **Proportional Representation:** Emphasizes the percentage contribution of each component to the total, rather than the absolute values.
- **Comparison:** Facilitates comparison of the *composition* of different groups or categories.
- **Part-to-Whole:** Clearly shows how individual parts contribute to the overall sum.
- **Fixed Height:** Each bar typically extends to the same height (representing 100%), making visual comparison of proportions easier.

Use:

- **Market Share Analysis:** Showing the percentage of market held by different companies over time.
- **Demographic Breakdown:** Illustrating the age or gender distribution within different population groups.

- **Survey Results:** Displaying the proportion of "agree," "disagree," and "neutral" responses for various questions.

4. Line Chart

It's a chart that shows trends in data over a continuous period or across ordered categories, primarily used to visualize how a data variable changes over time or another ordered sequence.

Properties:

- **Trend Visualization:** Excellent for showing how data evolves or trends over time.
- **Continuity:** Implies a continuous progression between data points, even if only sampled at discrete intervals.
- **Multiple Series:** Can display multiple data series simultaneously for comparison, each represented by a distinct line.
- **X-axis as Continuous Scale:** Typically, the X-axis (horizontal) represents a continuous variable like time, while the Y-axis (vertical) represents the measured value.

Use:

- **Time Series Data:** Tracking stock prices over months, temperature changes over a day, or sales figures over years.
- **Performance Monitoring:** Visualizing website traffic, system performance metrics, or production output over time.
- **Comparison of Trends:** Comparing the growth of different products or the performance of various marketing campaigns over the same period.
- **Forecasting:** Predicting future values based on past trends.

5. Area Chart

An area chart is a type of line chart where the space between the line and the X-axis is filled with colour or shading.

Properties:

- **Magnitude Emphasis:** The filled area visually represents the volume or total value of the data, making it easy to see the "amount" of change.
- **Trend Visualization:** Like line charts, they are excellent for showing trends and patterns over a continuous period.

- **Visual Impact:** The filled areas can make trends and comparisons more visually impactful than a simple line chart, especially for large changes.

Use:

- **Cumulative Totals:** Displaying cumulative sales over a year, total website traffic over time, or total revenue from different product lines.
- **Volume Changes:** Illustrating changes in volume, such as disk space usage, population growth, or resource consumption.
- **Highlighting Trends and Shifts:** Emphasizing the overall upward or downward movement of data and when significant shifts occur.

6. Stacked Area Chart

A stacked area chart is an extension of the area chart where multiple data series are "stacked" on top of each other.

Properties:

- **Part-to-Whole Relationship:** Clearly illustrates how individual components contribute to a changing overall total.
- **Cumulative Total:** The top edge of the uppermost layer represents the sum of all categories at any given point.
- **Trend and Magnitude:** Shows the trend of each individual category, as well as the overall trend and magnitude of the combined data.
- **Ordered Categories:** The order in which categories are stacked can influence readability; often, the largest or most stable category is placed at the bottom.

Use:

- **Composition Over Time:** Showing how the composition of a total changes, such as the breakdown of revenue by product line over quarters.
- **Resource Allocation:** Visualizing the distribution of resources (e.g., budget, time) across different activities over a project's duration.
- **Market Share Evolution:** Displaying the market share of various competitors over time, where the entire chart represents the total market.
- **Demographic Shifts:** Illustrating changes in the proportion of different age groups or ethnicities within a population over decades.

7. 100% Stacked Area Chart

A 100% stacked area chart is a variation of the stacked area chart where the total height of the chart at any given point is normalized to 100%.

Properties:

- **Relative Proportion Emphasis:** Primarily highlights the *percentage distribution* of categories, not their absolute values or the total sum.
- **Consistent Height:** The top edge of the chart remains flat, representing 100% at all times, making it easy to compare proportional changes.
- **Part-to-Whole Dynamic:** Clearly shows how the proportion of each part changes relative to the whole over the progression of time or sequence.
- **Trend of Proportions:** Illustrates the trends of each category's share of the total.

Use:

- **Market Share Analysis:** Visualizing changes in the percentage of market controlled by different companies over time.
- **Resource Allocation Proportions:** Showing how the percentage of a budget or time is allocated across different departments or tasks over a project's lifecycle.
- **Demographic Shifts:** Displaying how the percentage composition of different age groups or ethnic backgrounds within a population changes over decades.
- **Survey Response Proportions:** Illustrating the changing percentage of "agree," "disagree," or "neutral" responses to a question over time.

8. Line and stacked column chart

A line and stacked column chart is a hybrid visualization that combines a stacked column chart with one or more line charts on the same axes (or often with a secondary Y-axis). The stacked columns show the composition of totals, while the line(s) display trends of a different, but related, metric.

Properties:

- **Dual Perspective:** Allows for the simultaneous visualization of "part-to-whole" relationships (stacked columns) and "trends" (lines).
- **Correlation Insight:** Facilitates understanding of how the total composition or individual components relate to a different, trending metric.
- **Multiple Data Types:** Can effectively display data with different scales or units by utilizing a secondary Y-axis for the line series if necessary.
- **Hierarchical Information:** The stacked columns provide a detailed breakdown, while the line offers an overarching trend.

Use:

- **Sales and Profit Margins:** Stacked columns show sales breakdown by product, while a line shows the profit margin percentage over time.
- **Website Traffic and Conversion Rate:** Stacked columns show traffic sources (e.g., organic, paid) with a line representing the conversion rate.
- **Budget Allocation and Project Completion:** Stacked columns display budget spent by department, with a line showing the percentage of project completion.
- **Production Output and Defect Rate:** Stacked columns show the total units produced by shift, while a line indicates the defect rate

9. Line and clustered column chart

A line and clustered column chart is a combo chart that combines a clustered column chart with one or more line charts on the same visual. The clustered columns display comparisons of discrete values for multiple subcategories within each main category, while the line(s) show a trend of a related, continuous metric, often over time.

Properties:

- **Dual Comparison & Trend:** Allows for a quick visual comparison of multiple data series within categories (columns) and simultaneously tracks a trend of another related metric (line).
- **Clustered Grouping:** Columns are grouped together by a primary category, enabling comparison between subcategories *within* that group.
- **Different Scales (Optional):** Can effectively handle data with different units or scales by utilizing a secondary Y-axis for the line series, if needed.
- **Relationship Insight:** Helps to identify potential correlations or inverse relationships between categorical comparisons and the overall trend.

Use:

- **Sales vs. Customer Satisfaction:** Clustered columns showing sales volume for different product lines across various regions, with a line representing the average customer satisfaction score over those regions.
- **Production Output vs. Defect Rate:** Clustered columns displaying the number of units produced by different manufacturing shifts for various products, while a line shows the defect rate for each product/shift combination.

- **Budget Allocation vs. Project Progress:** Clustered columns illustrating budget spent by different departments on various project phases, with a line indicating the overall percentage of project completion over time.
- **Student Performance:** Clustered columns showing average scores in different subjects (e.g., Math, Science) for various classes, with a line representing the class average GPA.

10. Ribbon chart

A ribbon chart is a specialized visualization designed to show changes in ranking and value over time for different categories.

It's a type of chart that uses "ribbons" to connect categorical data points across a continuous axis (typically time).

Properties:

- **Rank Change Emphasis:** Its primary strength is clearly showing how the rank of different categories changes over the visualized period.
- **Value and Trend:** While showing rank, the thickness or height of the ribbons also indicates the magnitude of the value for each category.
- **Time-Series Data:** Ideal for data that changes over time, allowing for dynamic comparisons.
- **Visual Flow:** The connecting ribbons create a strong visual flow that makes it easy to follow individual categories and their shifts in position.

Use:

- **Market Share Evolution:** Tracking how the market share (and thus rank) of different companies or products changes over quarters or years.
- **Sales Performance Ranking:** Visualizing how the sales performance of various product lines or regions shifts in rank month by month.
- **Player/Team Rankings:** Showing the progression of sports team or individual player rankings over a season.
- **Website Traffic Sources:** Illustrating how the top traffic sources for a website change in their contribution over time.

11. Waterfall Chart

A waterfall chart is a specialized visualization that illustrates how an initial value is affected by a series of positive and negative changes, ultimately leading to a final value. It's often referred to as a "bridge chart" or "flying bricks chart" due to its appearance.

Properties:

- **Cumulative Effect:** Clearly visualizes the running total as values are added or subtracted.
- **Sequential Changes:** Presents changes in a step-by-step, chronological, or categorical order.
- **Positive/Negative Distinction:** Uses different colors (e.g., green for increases, red for decreases) to highlight positive and negative contributions.
- **Floating Columns:** Intermediate values are represented by "floating" columns, connecting the end of one segment to the start of the next.
- **Initial/Final Totals:** The starting and ending values are typically represented by full bars that begin or end at the zero baseline.
- **Subtotals:** Can include intermediate subtotal bars to break down complex progressions.

Use:

- **Financial Analysis:** Explaining profit and loss statements, showing how revenue, costs, and expenses lead to net profit.
- **Budget Variance:** Illustrating how actual spending deviates from a budget, highlighting contributions that caused the variance.
- **Inventory Analysis:** Tracking changes in inventory levels, showing additions from new stock and subtractions from sales.
- **Project Management:** Visualizing the progress of a project's cost, showing initial budget, expenditures, and remaining funds.
- **Performance Metrics:** Analysing changes in key performance indicators (KPIs) over time due to specific contributing factors.

12. Funnel

A funnel chart is a visualization that depicts the progressive reduction of data as it passes through sequential stages of a process.

Properties:

- **Sequential Process:** Designed specifically for processes with ordered, dependent stages where a quantity is expected to decrease from one stage to the next.
- **Attrition/Conversion Rate:** Clearly shows where data "drops off" or converts at each step, highlighting areas of efficiency or inefficiency.
- **Bottleneck Identification:** Makes it easy to visually pinpoint stages with significant drops, indicating potential bottlenecks.

- **Proportional Representation:** The size of each segment (width or height) is proportional to the value it represents, allowing for quick visual comparison of quantities at each stage.

Use:

- **Sales Pipeline Analysis:** Tracking prospects from initial contact to closed deals, identifying where leads are lost.
- **Website Conversion:** Visualizing user journeys from website visits to product views, adding to cart, and final purchase.
- **Recruitment Process:** Showing the flow of candidates from application submission to interviews, offers, and hiring.
- **Marketing Campaign Performance:** Analysing the effectiveness of campaigns by tracking metrics like email opens, clicks, and conversions.
- **Order Fulfilment:** Monitoring the progress of orders from receipt to processing, shipping, and delivery.

13. Scatter Chart

A scatter chart (also known as a scatter plot or scatter diagram) is a graphical representation that uses dots to display values for two different numerical variables.

Properties:

- **Relationship Visualization:** Primarily used to observe and show the relationship, correlation (positive, negative, or none), and patterns between two quantitative variables.
- **Individual Data Points:** Each point on the chart corresponds to a unique observation, allowing for the identification of outliers or clusters.
- **No Time Dependence:** Unlike line charts, the X-axis does not necessarily represent time; both axes are typically continuous numerical scales.
- **Non-linear Relationships:** Can reveal non-linear or complex relationships between variables that might not be evident in other chart types.
- **Third Variable (Optional):** Additional variables can be encoded using the size, shape, or color of the data points (making it a bubble chart if size is used).

Use:

- **Correlation Analysis:** Determining if there's a positive, negative, or no correlation between two variables (e.g., studying hours vs. test scores, temperature vs. ice cream sales).

- **Outlier Detection:** Identifying data points that deviate significantly from the general pattern, which may indicate anomalies or errors.
- **Trend Identification:** Visualizing patterns like linear or non-linear trends in large datasets.
- **Scientific and Statistical Analysis:** Widely used in fields like science, engineering, and statistics to explore hypotheses and understand relationships in experimental data.
- **Identifying Clusters:** Grouping data points that are close together to uncover distinct segments or behaviours within the data.

14. Pie Chart

A pie chart is a circular statistical graphic divided into slices to illustrate numerical proportion.

The entire circle represents the total sum of all categories (100%).

Properties:

- **Part-to-Whole Relationship:** Clearly shows how each category contributes to a single total.
- **Proportional Representation:** The size of each slice directly corresponds to its percentage of the whole.
- **Limited Categories:** Best suited for a small number of categories (typically 2-6) to maintain readability; too many slices become difficult to distinguish and compare.
- **No Trends or Changes:** Not suitable for showing changes over time or relationships between variables.

Use:

- **Market Share:** Displaying the percentage of market controlled by different companies.
- **Budget Allocation:** Showing how different categories contribute to a total budget.
- **Demographic Breakdown:** Illustrating the distribution of a population by age group, gender, or other categories.
- **Survey Responses (Proportions):** Presenting the percentage breakdown of responses to a single-choice question.

15. Donut Chart

A donut chart is a variation of the pie chart, with a blank center that allows for additional information or a clearer visual focus.

Properties:

- **Part-to-Whole Relationship:** Like a pie chart, it effectively shows how individual categories contribute to a single total.
- **Central Space for Information:** The empty center can be used to display summary statistics, a title, or a key metric related to the data, enhancing context.
- **Improved Readability for Multiple Series (Sometimes):** When displaying multiple donut charts, their central holes can make them less visually "heavy" than pie charts.
- **Arc Length for Proportion:** Proportions are represented by the length of the arc of each segment, rather than the area of a slice. This can sometimes make comparisons slightly easier than with pie charts.
- **Limited Categories:** Similar to pie charts, they are best suited for a small number of categories to maintain clarity.

Use:

- **Dashboard Summaries:** Ideal for dashboards where a key metric (e.g., total sales) can be displayed in the center while the surrounding donut shows its breakdown (e.g., sales by region).
- **Performance Tracking:** Showing completion rates or progress (e.g., 75% complete) in the center, with the donut representing the proportion of completed vs. remaining tasks.
- **Financial Allocation:** Visualizing how a budget is distributed, with the total budget amount in the center.
- **Survey Response Rates:** Displaying the percentage of different responses for a question, with the total response count in the middle.

16. Treemap

A treemap is a data visualization that displays hierarchical (tree-structured) data as a set of nested rectangles.

The size (area) of each rectangle is proportional to a quantitative value, and often, color is used to represent another quantitative or categorical variable.

Properties:

- **Hierarchical Representation:** Excellently visualizes data with parent-child relationships across multiple levels.
- **Space-Efficient:** Can display a large amount of hierarchical data in a compact space, making it efficient for dashboards or reports with limited real estate.

- **Proportional Sizing:** The area of each rectangle directly corresponds to its assigned numerical value, allowing for quick visual comparison of magnitudes.
- **Color Encoding:** Colors can be used to add another layer of information, such as categorizing by a different attribute or showing a secondary measure (e.g., performance, status).
- **Outlier/Pattern Detection:** Helps in quickly identifying the largest contributors, smallest elements, or specific patterns within the hierarchical structure.

Use:

- **File System Visualization:** Showing disk space usage, where larger rectangles represent folders and smaller ones represent subfolders or files.
- **Market Capitalization:** Displaying the relative sizes of different industries or companies within a market, with sub-rectangles showing individual companies.
- **Budget Allocation:** Visualizing how a budget is distributed across main departments and then further broken down by sub-departments or cost centers.
- **Sales Performance by Product Category:** Showing total sales by major product categories, and then by individual products within those categories.
- **Population Demographics:** Representing population distribution by continent, country, and then by region or city, with size indicating population count.

17. Map

This is the simplest map visual in Power BI, primarily used to plot individual geographic data points (like cities, countries, or precise latitude/longitude coordinates) as bubbles or markers on a world map. It leverages Bing Maps for geocoding.

Properties:

- **Point-based:** Displays data as distinct points (bubbles or circles) on a map.
- **Size Encoding:** The size of the bubbles can be used to represent a numerical measure (e.g., larger bubbles for higher sales).
- **Color Encoding:** The color of the bubbles can be used to categorize data or show another measure.
- **Basic Geo-coding:** Automatically translates geographic names (like "London" or "USA") into map coordinates, though direct latitude/longitude provides more accuracy.
- **Interactive:** Supports zoom, pan, and tooltips on hover.

Use:

- **Location of Customers/Stores:** Showing the geographic distribution of customers, retail outlets, or service centers.
- **Event Locations:** Plotting the locations of incidents, natural disasters, or points of interest.
- **Distribution of Resources:** Visualizing where assets or personnel are located.

18. Filled Map

A filled map, also known as a choropleth map, uses color shading or intensity to display data values across predefined geographical regions (e.g., countries, states, provinces, or postal codes). The color saturation reflects the magnitude of the data.

Properties:

- **Region-based:** Fills entire geographical areas with color based on a data measure.
- **Color Intensity for Value:** Darker or more intense colors typically indicate higher values, while lighter colors indicate lower values, making it easy to spot concentrations.
- **Predefined Regions:** Works best with standard geographic boundaries that Power BI (via Bing Maps) can recognize.
- **Comparison Across Regions:** Excellent for comparing data values across different regions.

Use:

- **Regional Sales Performance:** Identifying top-performing or underperforming sales territories.
- **Demographic Distribution:** Visualizing population density, income levels, or other demographic data by region.
- **Disease Incidence:** Showing the prevalence of a disease across different states or countries.

19. Azure Map

It's a Power BI visual that allows users to overlay multiple layers of location-based data (like points, bubbles, heatmaps, filled regions, and even custom GeoJSON shapes) onto a dynamic and interactive map canvas powered by Azure Maps.

Properties:

- **Multi-layered:** Supports various data layers (Bubble, 3D Column, Filled Map, Heatmap, Cluster, Path, Reference, Tile, Traffic) on a single map, allowing for complex geospatial analysis.
- **Real-time Data (with Azure):** When connected to Azure services, it can incorporate real-time data like traffic conditions.
- **Customization:** Provides extensive formatting options for map styles, colors, sizes, and borders of visual elements.
- **Scalability:** Designed to handle larger datasets and complex mapping scenarios efficiently.

Use:

- **Advanced Location Intelligence:** Gaining deeper insights into business data's location context (e.g., how customer distribution affects sales).
- **Sales Territory Analysis:** Analyzing sales performance across complex, custom-defined territories, identifying high-performing or underserved areas.
- **Urban Planning & Public Services:** Visualizing infrastructure, service coverage, incident locations, and resource allocation within cities or regions.
- **Environmental Monitoring:** Mapping environmental data, such as pollution levels, weather patterns, or resource distribution.

20. Gauge

A Gauge visual is a single-value visualization that displays progress toward a specific goal or target.

Properties:

- **Single Value Focus:** Designed to highlight one specific metric and its current state.
- **Progress Toward Goal:** Effectively illustrates how close the current value is to a desired target.
- **Visual Range:** Typically includes a minimum, maximum, and often colored ranges (e.g., red, yellow, green) to categorize performance (e.g., poor, acceptable, good).
- **Target Indicator:** A distinct line or "needle" on the gauge indicates the target value, providing an immediate visual benchmark.

Use:

- **KPI Monitoring:** Tracking key performance indicators like sales targets, customer satisfaction scores, project completion rates, or production quotas.

- **Performance Assessment:** Quickly determining if a metric is meeting, exceeding, or falling short of its goal.
- **Dashboard Summaries:** Providing a high-level overview of critical metrics on executive dashboards.
- **Individual Performance:** Showing an individual's progress against their personal goals or metrics.

21. Card

A **Card visual** is a simple yet powerful visualization used to display a single, prominent numeric value. It's a compact visual element that presents a single data point (a measure or a key performance indicator) in a large, easily readable format, often accompanied by a label.

Properties:

- **Single Value Focus:** Designed to highlight one important number, making it immediately visible and understandable.
- **Minimalist Design:** It has a clean, uncluttered layout, focusing solely on the number.
- **Filtering Capability:** Reacts to filters from other visuals, dynamically updating the displayed value.

Use:

- **Key Performance Indicators (KPIs):** Displaying critical metrics like total sales, average profit, current inventory count, or number of active users.
- **Dashboard Summaries:** Providing high-level snapshots of performance on a dashboard.
- **Quick Overviews:** Giving users an immediate understanding of the current status of a particular metric.
- **Comparing Single Values:** When multiple card visuals are used side-by-side, they allow for quick comparison of different key metrics.

22. Multi Row Card

It's a visual that serves as a list of fields, where each row (or "card") typically represents a distinct item or category, and multiple values associated with that item are displayed vertically beneath it.

Properties:

- **List-like Display:** Presents data in a stacked, list-like format, making it easy to scan multiple related values for several items.
- **Multiple Fields per Row:** Unlike a single Card visual, it can show several measures and/or categorical fields for each row of data.
- **Row-wise Organization:** Each "card" or section within the visual corresponds to a row in your dataset, grouping related information.
- **Filtering Capability:** It is interactive and responds to filters applied from other visuals in the report.
- **Compact Information Display:** Efficiently displays a summary of multiple data points for several categories in a relatively small space.

Use:

- **Top N Lists:** Displaying the top (or bottom) N customers, products, or regions along with their key metrics (e.g., Customer Name, Total Sales, Profit Margin).
- **Summary of Key Entities:** Showing a concise summary for a selected set of entities (e.g., for selected stores, list their Manager, Address, and Total Sales).
- **Drill-through Reports:** When a user drills through from another visual, a multi-row card can provide a detailed summary of the selected item's attributes and metrics.
- **Inventory or Asset Listings:** Displaying details for a list of items such as Product Name, Stock Level, Last Order Date, and Price.

23. KPI

A **Key Performance Indicator (KPI) visual in Power BI** is a specialized chart designed to quickly communicate the progress of a measurable goal.

It's a visual cue that displays a primary "indicator" value, its "target goal," and a "trend axis" (often a mini-chart in the background), indicating whether the performance is meeting, exceeding, or falling short of the target, along with its progression over time.

Properties:

- **Goal-Oriented:** Its core purpose is to show performance against a predefined target.
- **Three Key Elements:** Requires a **base measure** (the actual value), a **target measure/value** (the goal), and a **trend axis** (typically a date/time field to show progression).
- **Visual Status Cues:** Uses color-coding (e.g., green for good, red for bad) and icons (e.g., checkmark, exclamation point) to instantly convey the status relative to the target.

- **Trend Visualization:** Includes a small line or area chart in the background to show the historical trend of the base measure.
- **Distance to Goal:** Can display the absolute difference or percentage difference from the target.
- **Directionality:** Allows defining whether a higher value is "good" or "bad" (e.g., higher sales are good, higher defect rates are bad).

Use:

- **Monitoring Business Objectives:** Tracking how well an organization is achieving its strategic, financial, or operational goals (e.g., "Sales vs. Target," "Customer Satisfaction Score," "On-Time Delivery Rate").
- **Performance Management:** Quickly assessing if specific metrics are on track, flagging areas that require immediate attention or intervention.
- **Executive Dashboards:** Providing a concise, high-level overview of critical business health indicators.
- **Identifying Trends and Variances:** Observing both the current status and the underlying trend to understand performance trajectories and variances from goals.

24. Slicer

A **Slicer** is an on-report visual filter that allows users to interactively filter data displayed in other visuals on the same page.

It's a standalone filtering mechanism presented directly on the report canvas, enabling users to select specific values or ranges from a field to narrow down the data shown in connected visualizations, without needing to open filter panes.

Properties:

- **Interactive Filtering:** Provides a user-friendly way for report consumers to filter data by simply clicking or selecting options directly on the report.
- **On-Report Presence:** Slicers are visible elements on the report page, making filtering capabilities immediately apparent.
- **Multiple Formats:** Can display filter options in various ways:
 - **List:** A vertical list of distinct values (e.g., product names).
 - **Dropdown:** A compact dropdown menu for selections.
 - **Between/Range:** For numerical or date fields, it allows selection of a range using sliders or input boxes.
 - **Relative Date/Time:** Filters by relative periods (e.g., "last 3 months," "next year").

- **Cross-filtering:** By default, a slicer filters all other visuals on the report page that are based on the same data model. Its interactions can be managed.

Use:

- **User-driven Data Exploration:** Empowering end-users to customize the data view to their specific interests without modifying the report's underlying structure.
- **Dynamic Reports:** Creating dynamic and interactive dashboards where users can quickly switch between different segments of data (e.g., view sales by specific region, year, or product category).
- **Segment Analysis:** Focusing on particular segments of data for deeper analysis (e.g., filtering for only "online sales" or "customers in North America").

25. Table

A table visual in Power BI displays data in a row-and-column format, similar to a spreadsheet. It presents exact numerical or textual values, allowing users to see detailed, granular data.

Properties:

- **Exact Values:** Shows precise data points, unlike charts that summarize or visualize trends.
- **Sorting:** Users can sort the table by any column in ascending or descending order.
- **Scrollable:** Can display large datasets, with scrollbars enabling navigation through all rows.
- **Conditional Formatting:** Supports applying rules to format cells based on their values (e.g., color scales, data bars, icons) to highlight trends or exceptions.
- **Interactive:** Responds to filters and slicers from other visuals and can cross-filter other visuals when a row or column is selected.
- **Totals/Subtotals (Optional):** Can display totals at the bottom for numeric columns.

Use:

- **Detailed Data Review:** For examining specific transactions, records, or line items (e.g., individual sales orders, customer details, product specifications).
- **Comparing Exact Figures:** When precise numerical comparisons are required across different categories (e.g., comparing sales figures for each product in different regions).

- **Exporting Data:** Users can often export the underlying data from a table visual for further analysis outside Power BI.

26. Matrix

A matrix visual in Power BI is a grid-like display that aggregates and summarizes data across two or more dimensions (rows and columns), similar to a pivot table in Excel. It allows for hierarchical drilling down into data.

Properties:

- **Hierarchical Display:** Supports multiple fields in rows and columns, allowing for drilling down and expanding/collapsing levels to view data at different granularities.
- **Cross-Tabulation:** Aggregates data at the intersection of selected row and column headers, making it excellent for cross-tabulation of measures across multiple categories.
- **Subtotals & Totals:** Automatically calculates and displays subtotals for each level of the hierarchy and grand totals for rows and columns.
- **Interactive:** Responds to filters and slicers from other visuals and can cross-filter other visuals when a cell, row, or column header is selected.
- **Concise Summarization:** Efficiently summarizes large amounts of data in a compact, readable format.

Use:

- **Multi-dimensional Analysis:** Analyzing sales by product category *and* by region, or expenses by department *and* by month.
- **Financial Reporting:** Creating income statements, balance sheets, or detailed budget versus actual reports with drill-down capabilities.
- **Performance Dashboards:** Summarizing key metrics across various dimensions (e.g., revenue by product and customer segment).
- **Comparing Aggregated Data:** When you need to compare aggregated values across multiple categories and subcategories in a structured grid format.
- **Exploring Data Hierarchies:** Allowing users to expand and collapse categories to explore data from a high-level summary down to granular details.

27. R script visual

An R script visual in Power BI allows users to create custom visualizations by writing and executing R code directly within Power BI Desktop. It leverages Power BI's data modeling capabilities and R's extensive statistical graphics packages.

Properties:

- **Custom Visualization:** Enables the creation of highly customized and advanced chart types not natively available in Power BI.
- **Statistical Analysis Integration:** Seamlessly integrates R's powerful statistical computing and graphical capabilities (e.g., ggplot2, lattice).
- **Data Input:** Automatically feeds data from Power BI fields into an R dataframe, which can then be manipulated and visualized by the R script.
- **Image Output:** The R script generates a plot image that Power BI displays as a static visual on the report canvas.
- **Interactivity (Limited):** While the R script itself is static, the visual responds to filtering and cross-highlighting from other Power BI visuals, rerunning the R script with the filtered data.
- **R Environment Dependency:** Requires an R distribution installed on the machine running Power BI Desktop and appropriate R packages.

Use:

- **Advanced Statistical Charts:** Creating specialized plots like density plots, violin plots, network graphs, correlation matrices, or highly customized scatter plots with regression lines.
- **Predictive Analytics Visualizations:** Visualizing outputs from R-based predictive models, such as forecast confidence intervals or anomaly detection plots.
- **Data Exploration:** Performing complex statistical transformations and visualizing the results when Power BI's native capabilities are insufficient.
- **Unique Branding/Design:** Generating visualizations with very specific aesthetic requirements or branding elements using R's flexible plotting options.
- **Academic/Research Reports:** When specific statistical graphics common in academic fields are required in a Power BI report.

28. Python visual

A Python visual in Power BI allows users to create custom visualizations by writing and executing Python code directly within Power BI Desktop. It leverages Power BI's data and Python's extensive libraries for data manipulation and visualization.

Properties:

- **Custom Visualization:** Enables the creation of highly customized and advanced chart types not natively available in Power BI, using libraries like Matplotlib, Seaborn, Plotly, etc.
- **Data Science Integration:** Seamlessly integrates Python's powerful capabilities for data cleaning, transformation, statistical analysis, and machine learning outputs.
- **Data Input:** Automatically feeds data from Power BI fields into a Pandas Data Frame, which can then be manipulated and visualized by the Python script.
- **Image Output:** The Python script generates a plot image that Power BI displays as a static visual on the report canvas.
- **Interactivity (Limited):** While the Python visual itself is typically a static image, it responds to filtering and cross-highlighting from other Power BI visuals, causing the Python script to re-execute with the filtered data.
- **Python Environment Dependency:** Requires a Python distribution installed on the machine running Power BI Desktop and appropriate Python packages.

Use:

- **Advanced Statistical Charts:** Creating specialized plots like complex scatter plots with multiple regression lines, violin plots, ridge plots, or highly customized dashboards layouts using Python's plotting libraries.
- **Machine Learning Model Outputs:** Visualizing the results of Python-based machine learning models, such as cluster visualizations, classification boundaries, or anomaly detection plots.
- **Complex Data Transformations & Visualizations:** When Power BI's native data transformation or visualization capabilities are insufficient for specific analytical needs.
- **Unique Aesthetic Requirements:** Generating visualizations with very specific aesthetic or branding requirements that are easier to achieve with Python's granular control over plot elements.
- **Integration with Python Ecosystem:** Leveraging existing Python scripts or data science workflows directly within Power BI for richer analytics and visualization.

29. Key Influencers

A **Key Influencer** is an AI-powered visualization that automatically analyzes your data to identify the factors (or "influencers") that drive a specific metric or outcome you are interested in.

It uses machine learning algorithms (like logistic regression for categorical targets and linear regression for continuous targets) to discover and rank the variables and their values that have the most significant statistical impact on a chosen target metric.

Properties:

- **AI-Driven Insights:** Automatically uncovers hidden patterns and relationships in your data, acting as a "root cause analysis" tool.
- **Target Metric Analysis:** Requires one "Analyze" field (the outcome you want to understand) and multiple "Explain by" fields (potential factors that might influence the outcome).
- **Categorical & Continuous Targets:** Can analyze what influences a categorical outcome (e.g., "why did customer churn happen?") or a continuous outcome (e.g., "what influenced sales to increase?").
- **Two Views:**
 - **Key Influencers:** Shows the top individual factors (e.g., "if Customer Segment is 'New', it's X times more likely to lead to Y outcome").
 - **Top Segments:** Identifies groups of data points (combinations of multiple factors) that significantly contribute to the target outcome.
- **Interactive Explanations:** Provides natural language summaries and supporting visuals (like column charts or scatter plots) to explain *why* a factor is an influencer.
- **Relative Importance:** Ranks factors by their statistical impact, making it easy to see which ones matter most.
- **Dynamic:** The analysis re-runs and updates as filters or slicers are applied to the report.

Use:

- **Understanding Churn/Attrition:** Identifying factors that lead to customer churn, employee turnover, or subscription cancellations.
- **Improving Customer Satisfaction:** Discovering what influences high or low customer satisfaction ratings.
- **Boosting Sales/Revenue:** Pinpointing factors that drive increased sales, higher average order value, or conversion rates.
- **Diagnosing Performance Issues:** Uncovering the root causes behind declining performance metrics in any area (e.g., production defects, website bounce rates).
- **Strategic Decision-Making:** Providing data-backed insights to guide business strategies, prioritize initiatives, and optimize processes.

30. Decomposition tree

A Decomposition tree is an AI-driven visualization in Power BI that allows for interactive "root cause" analysis by breaking down a high-level measure into its

contributing dimensions across multiple levels. It enables users to explore how a total is composed across various attributes.

Properties:

- **AI-Driven Exploration:** Users can choose to let the AI automatically find the next path to break down a measure ("High value" or "Low value"), or they can manually select the dimensions.
- **Hierarchical Breakdown:** Allows for drilling down into multiple levels of dimensions to understand the components of a selected measure.
- **Interactive Analysis:** Users can interactively expand and collapse branches of the tree, exploring different paths of decomposition.
- **Two Input Fields:** Requires one "Analyze" field (the measure you want to break down) and multiple "Explain by" fields (the dimensions you want to break it down by).
- **Visual Path:** Clearly shows the flow from the highest-level total down to more granular components through connecting nodes.
- **Contribution View:** Each node typically shows the value and sometimes the percentage contribution of that segment to its parent.

Use:

- **Root Cause Analysis:** Understanding why a particular sales figure is high or low by breaking it down by product, region, customer segment, etc.
- **Profitability Analysis:** Decomposing total profit by product line, sales channel, or customer type to identify key drivers.
- **Expense Analysis:** Breaking down total expenses by department, cost center, or expense category to find areas of high spending.
- **Performance Deep Dive:** Exploring the components of any aggregated metric (e.g., website traffic, production defects) to pinpoint specific contributing factors.
- **Ad-hoc Exploration:** Providing an intuitive way for business users to self-serve and explore complex relationships in their data without needing pre-defined drill-downs.

31. Q&A

The Q&A visual (and feature) in Power BI allows users to ask natural language questions about their data, and Power BI automatically generates appropriate visualizations or answers in response.

Properties:

- **Natural Language Processing (NLP):** Understands questions typed in plain English (or other supported languages), translating them into queries against the underlying data model.
- **Automatic Visualization:** Dynamically selects and renders the most suitable chart type (e.g., bar chart, line chart, table, card) based on the question asked.
- **Interactive:** Users can refine their questions, and the visual updates in real-time. The generated visuals are interactive like other Power BI visuals.
- **Synonym & Term Learning:** Can be trained by report designers to understand specific business jargon or synonyms for fields, enhancing its accuracy.
- **Pre-defined Questions:** Report creators can suggest popular questions to guide users.
- **Leverages Data Model:** Relies heavily on a well-structured data model with proper field naming and relationships for accurate responses.

Use:

- **Ad-hoc Data Exploration:** Enabling business users to quickly get answers to their specific questions without needing to build reports or understand complex data models.
- **Self-Service Analytics:** Empowering non-technical users to perform on-the-fly analysis by simply typing what they want to know.
- **Quick Insights:** Rapidly generating visualizations for quick insights during meetings or presentations.
- **Discovering New Questions:** By seeing what users are asking, report creators can identify common analytical needs and build more targeted reports.
- **Accessibility:** Making data analysis more accessible to a broader audience who may not be familiar with traditional BI tools.

32. Narrative

It's a text box on the report canvas that uses artificial intelligence to interpret and explain the key findings, trends, and outliers within your data, providing an automated "story" or commentary that updates dynamically with filters and interactions.

Properties:

- **AI-Generated Text:** Automatically creates human-readable summaries and explanations of data patterns.
- **Dynamic Updates:** The narrative text and any embedded values automatically refresh as users apply filters, drill down, or cross-filter other visuals.

- **Contextual Insights:** Can summarize an entire report page or focus on specific visuals, providing relevant insights like trend changes, top/bottom performers, or key drivers.
- **Customization:** While AI-generated, the text is fully editable. Users can add their own prose, format the text, and even embed dynamic values (measures or fields) directly into the narrative using natural language.
- **Highlighting Key Data:** Often underlines or bolds numerical values and key terms derived from the data model, making them stand out.
- **Copilot Integration (Newer):** With Copilot enabled in Fabric, the narrative visual can leverage more advanced AI to generate executive summaries, bulleted insights, and more sophisticated explanations.

Use:

- **Automated Data Storytelling:** Quickly generating written summaries for reports, saving time for report creators and making data more accessible to report consumers.
- **Highlighting Key Takeaways:** Drawing attention to the most important insights, trends, and anomalies in a report, ensuring stakeholders grasp critical information immediately.
- **Executive Summaries:** Providing a concise, automatically updated narrative summary on a dashboard for quick understanding of overall performance.
- **Explaining Complex Data:** Making complex data visualizations and underlying patterns easier to understand for a broader audience, especially those less familiar with data analytics.
- **Accelerating Report Creation:** Reducing the manual effort required to write textual interpretations of data, allowing report developers to focus on visual design and data modelling.

33. Goals

Goals (formerly called Metrics in Power BI) are a feature within Power BI that allows organizations to track, monitor, and manage key business objectives against targets in a dedicated, scorecard-like view. They provide a high-level summary of performance and enable collaboration around strategic goals.

Properties:

- **Performance Tracking:** Designed to monitor progress against specific business targets over time.
- **Scorecard View:** Presents multiple goals in a consolidated, easy-to-digest scorecard format.

- **Status & Trend:** Each goal typically displays its current value, target, status (e.g., On track, At risk, Behind), and a trend indicator (e.g., up, down, flat).
- **Connected to Data:** Goals are linked to underlying Power BI reports and semantic models, meaning their values update automatically as the data refreshes.
- **Check-ins & Notes:** Supports manual "check-ins" where users can update progress, add notes, and adjust the status of a goal, fostering collaboration.
- **Ownership:** Goals can be assigned owners, promoting accountability.
- **Rules & Automation:** Statuses can be defined manually or automated based on rules tied to the data.
- **Hierarchical Organization:** Goals can be organized into scorecards and can support sub-goals for more detailed tracking.

Use:

- **Strategic Planning & Execution:** Monitoring the progress of company-wide or departmental strategic objectives.
- **Performance Management:** Tracking key performance indicators (KPIs) and their targets across different teams or individuals.
- **Executive Dashboards:** Providing a high-level overview of business health and progress towards critical objectives for leadership.
- **Team Collaboration:** Facilitating discussion and updates around shared goals, with the ability to add notes and insights directly within the platform.
- **Identifying Gaps:** Quickly pinpointing areas where performance is falling short of targets, enabling timely interventions.

34. Paginated report

It's a report type, often created using Power BI Report Builder (a separate tool), that adheres to a fixed layout, ensuring consistent formatting, page breaks, headers, and footers when rendered to print or export formats like PDF, Word, or Excel. They are based on the SQL Server Reporting Services (SSRS) Report Definition Language (RDL).

Properties:

- **Print-Optimized:** Designed to look exactly the same when printed or exported as a document (e.g., PDF) as they do on screen.
- **Pixel-Perfect Layout Control:** Offers precise control over element placement, sizing, and pagination, including repeating headers/footers on every page.
- **Multi-Page Support:** Can display entire tables or matrices with thousands of rows, spanning across numerous pages automatically.
- **Data-Driven Pages:** Can be configured to generate a new page for each unique value in a specific field (e.g., one invoice per customer).

- **Parameters:** Supports dynamic filtering of the report data based on user-selected parameters at runtime.
- **Limited Interactivity:** Less interactive than standard Power BI reports; typically used for viewing and printing rather than deep, ad-hoc data exploration.
- **Separate Authoring Tool:** Primarily created and edited using Power BI Report Builder, a standalone application, though they can connect to Power BI datasets.

Use:

- **Operational Reports:** Generating invoices, purchase orders, shipping labels, financial statements, or regulatory filings that require exact, fixed layouts.
- **Mass Distribution:** Creating reports for large-scale printing or automated distribution (e.g., emailing monthly statements to hundreds of customers).
- **Compliance & Audit:** Producing detailed audit trails, compliance reports, or legal documents where data integrity and consistent formatting are critical.
- **Long Tabular Reports:** When you need to print or export comprehensive lists or tables that contain a high volume of rows and need to flow seamlessly across pages.
- **Legacy SSRS Migration:** For organizations transitioning from SQL Server Reporting Services (SSRS), as paginated reports are built on the same underlying RDL technology.

35. Card (new)

The "Card (new)" in Power BI is an updated and enhanced version of the traditional Card visual, offering significantly more flexibility and customization options. It was introduced to address limitations of the old card, particularly the inability to display multiple metrics within a single visual container easily.

Properties:

- **Multiple Metrics in One Visual:** The most significant enhancement is the ability to add multiple data fields, each becoming its own "card" within the single (new) Card visual.
- **Reference Labels:** Allows for the addition of secondary "reference labels" for each metric, providing extra context like target values, percentage changes, or units.
- **Images & Icons:** Can incorporate images or icons alongside values, enhancing visual appeal and quick understanding.
- **Shape Customization:** Offers options to customize the shape of individual cards, including rounded corners or snipped corners.

- **Improved Performance & Accessibility:** Designed from the ground up for better rendering performance and improved accessibility (e.g., for screen readers) compared to overlapping multiple old card visuals.
- **Small Multiples Integration (evolving feature):** Increasingly supports small multiples capabilities, allowing you to break down the set of cards by a chosen dimension.

Use:

- **Comprehensive KPI Dashboards:** Creating a concise dashboard section that displays multiple critical KPIs (e.g., Total Sales, Total Profit, Average Order Value, Customer Count) in a single, well-formatted block.
- **Comparative Performance Overview:** Presenting key metrics for different categories side-by-side within one visual (e.g., "Sales This Year" vs. "Sales Last Year" vs. "Sales Target").
- **Detailed Financial Summaries:** Showcasing various financial figures (e.g., Revenue, COGS, Gross Profit, Operating Expenses) in a structured, easily digestible format.
- **Any Scenario Requiring Multiple Prominent Numbers:** Whenever a report needs to highlight several important individual numbers, the new card visual provides a clean, efficient, and highly customizable solution.
- **Dynamic Information Displays:** Using its formatting capabilities to create highly branded or visually appealing displays of key data points that automatically update with filters.

36. Button slicer

A button slicer in Power BI is a standard slicer visual that has been formatted to display its filter options as interactive buttons instead of a traditional list or dropdown. This style is often chosen for its clean appearance and ease of use, particularly for single-select or distinct categorical filters.

Properties:

- **Visual Appeal:** Offers a modern and intuitive interface compared to a standard list slicer, making it more aesthetically pleasing for dashboards.
- **Interactive Filtering:** Functions identically to a regular slicer, allowing users to click buttons to filter other visuals on the report page.
- **Compactness (Horizontal):** When configured horizontally, it can save vertical space on the report canvas.
- **Customizable States:** Buttons can be formatted to show different styles (e.g., color, border, font) for default, hover, pressed, and selected states.

- **Single or Multi-Select:** Can be set to allow users to select only one button at a time (single-select) or multiple buttons (multi-select), just like a regular slicer.
- **Search Bar (Optional):** Can include a search bar if there are many unique values to filter.

Use:

- **Filtering by Discrete Categories:** Ideal for filtering by categories with a limited number of distinct values, such as "Year," "Quarter," "Region," "Product Category," or "Gender."
- **Switching Views:** Providing quick options for users to switch between different data views or analyses based on a categorical selection.
- **Date Part Filtering:** Enabling users to quickly select specific years, months, or days of the week.
- **"Top N" Selection:** When integrated with filtering techniques, buttons can allow users to select "Top 5," "Top 10," etc.
- **Improving User Experience:** Enhancing the visual appeal and user-friendliness of reports by making filters more discoverable and clickable.

37. ArcGIS

It's a custom Power BI visual provided by Esri that allows users to create rich, interactive maps by visualizing Power BI data on a geospatial canvas. It enables the use of various map layers (points, heatmaps, boundaries), access to demographic data, and performance of spatial analysis directly within the report.

Properties:

- **Advanced Mapping Layers:** Supports plotting data as points, clusters, heatmaps, and filled (choropleth) maps, with options for 3D column representation.
- **Authoritative Basemaps & Reference Layers:** Provides access to a wide range of Esri's basemaps (e.g., street, satellite, topographic) and allows overlaying authoritative demographic data or custom geographic layers from ArcGIS Online or ArcGIS Enterprise.
- **Spatial Analysis Tools:** Offers built-in tools for basic spatial analysis like:
 - **Buffer/Drive Time:** Identifying areas within a certain distance or travel time from selected locations.
 - **Find Similar Locations:** Discovering locations that share similar attributes.
 - **Infographics:** Displaying detailed demographic information for selected areas.

- **Geocoding:** Converts addresses or place names into geographic coordinates for plotting on the map.
- **Interactive & Bi-directional Filtering:** Maps are interactive (zoom, pan, select) and can cross-filter other visuals in the report, and vice-versa.
- **Account Tiers:** Offers a standard (free) experience with limited functionality and a premium experience upon signing in with an ArcGIS organizational account, unlocking more features and content.

Use:

- **Enhanced Location Intelligence:** Gaining deeper, more nuanced insights from location-based data by integrating business data with rich geographic context.
- **Site Selection & Planning:** Identifying optimal locations for new stores, facilities, or service areas based on demographic data and spatial analysis.
- **Sales Territory Optimization:** Analyzing sales performance, customer density, and market potential within specific geographic territories.
- **Risk Assessment & Emergency Management:** Mapping incidents, resource distribution, and vulnerable areas for better planning and response.
- **Demographic and Market Segmentation:** Understanding the distribution and characteristics of target audiences or market segments across regions.
- **Infrastructure Management:** Visualizing assets, maintenance schedules, and service coverage on a map for utilities, telecommunications, etc.

38. Power Apps

The Power Apps visual in Power BI allows you to embed a fully interactive Power Apps canvas app directly into your Power BI report. This integration enables users to interact with and even *write back* data from within the Power BI report, bridging the gap between data consumption and data action.

Properties:

- **Bi-directional Data Flow:**
 - **Power BI to Power Apps:** Power BI data (from selected fields) can be passed as context to the embedded Power App. The app automatically receives the filtered data from the Power BI report.
 - **Power Apps to Power BI/Data Source:** Users can interact with the app to update data directly in the underlying data source or trigger other actions, potentially refreshing the Power BI report.
- **Interactive Controls:** Leverages the full range of Power Apps controls (buttons, text inputs, forms, galleries, etc.) for highly customized user interaction.

- **Customizable User Interface:** The embedded Power App provides a highly tailored user interface for specific business processes or data entry.
- **Contextual Actions:** Allows users to take action on specific data points shown in the Power BI report (e.g., select a customer in Power BI, then update their contact details directly in the embedded app).
- **Security & Permissions:** Inherits security from Power Apps and the underlying data sources it connects to.

Use:

- **Data Write-back:** Enabling users to update database records, add comments, or change statuses directly from a Power BI report (e.g., update a sales lead status, add notes to a customer record).
- **Scenario Planning/What-if Analysis:** Building simple input forms in Power Apps to adjust parameters that then feed back into calculations displayed in Power BI.
- **Workflow Integration:** Triggering workflows or business processes based on insights from Power BI data (e.g., initiating an approval process for a budget item).
- **Detailed Data Entry:** Allowing users to view summarized data in Power BI and then drill down to enter or edit granular details through the embedded app.
- **Personalized Experiences:** Creating highly customized data entry or interaction experiences tailored to specific user roles or business needs directly within the reporting environment.

39. Power Automate

The **Power Automate visual in Power BI** is a specialized button visual that allows users to trigger automated workflows (flows) built with Microsoft Power Automate directly from a Power BI report. This integrates data insights with actionable processes.

This flow can be designed to perform various automated tasks, often using data from the Power BI report as input.

Properties:

- **Actionable Insights:** Bridges the gap between viewing data and taking action on that data.
- **Trigger for Flows:** Serves as a manual trigger for an Instant Cloud Flow in Power Automate.
- **Data Contextual:** Can pass selected or filtered data from the Power BI report as dynamic inputs to the Power Automate flow. This means the flow's actions can be specific to the data the user is viewing.

- **Customizable Button:** The visual itself is a customizable button (text, color, size) that users click to initiate the flow.
- **Integration with Microsoft Ecosystem:** Seamlessly connects Power BI with hundreds of other applications and services through Power Automate's vast connector library (e.g., SharePoint, Outlook, Teams, Salesforce, Twitter).
- **Permissions Required:** The user running the flow needs the necessary permissions to execute the Power Automate flow.

Use:

- **Automated Notifications:** Sending emails or Teams messages with report insights or filtered data to stakeholders (e.g., "Send email to sales manager with Q1 sales report for selected region").
- **Data Write-back/Updates:** Updating records in external systems (like a CRM, SharePoint list, or database) based on data selections in Power BI (e.g., "Mark selected orders as 'Reviewed'").
- **Approval Workflows:** Initiating approval processes directly from a report (e.g., "Request approval for budget variance").
- **Report Distribution:** Automating the export and distribution of Power BI reports or underlying data to specific recipients or storage locations.
- **Task Management:** Creating new tasks in a project management tool (e.g., Planner, Azure DevOps) based on insights from a Power BI report.